

7.2.29. Define $H: \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R} \times \mathbb{R}$ as follows:

$$H(x, y) = (x + 1, 2 - y) \text{ for every } (x, y) \in \mathbb{R} \times \mathbb{R}.$$

- a. Is H injective? Prove or give a counterexample.
- b. Is H surjective? Prove or give a counterexample.

a. Is H injective? Prove or give a counterexample.

Proof:

$$\text{Suppose } H(x, y) = (x + 1, 2 - y).$$

We must show whether H is injective or not.

Definition of injective:

Outputs cannot be shared

Let $(x_1, y_1), (x_2, y_2) \in \mathbb{R} \times \mathbb{R}$

Suppose $H(x_1, y_1) = H(x_2, y_2)$, then $x_1 + 1 = x_2 + 1$
and $2 - y_1 = 2 - y_2$.

$$x_1 + 1 = x_2 + 1$$

$$(x_1 + 1) - 1 = (x_2 + 1) - 1$$

$$x_1 = x_2$$

$$2 - y_1 = 2 - y_2$$

$$(2 - y_1) - 2 = (2 - y_2) - 2$$

$$-y_1 = -y_2$$

$$y_1 = y_2$$

Hence, $x_1 = x_2$ and $y_1 = y_2$. Therefore, $(x_1, y_1) = (x_2, y_2)$,
which is what was to be shown.

□

Solution: H is injective.

b. Is H surjective? Prove or give a counterexample.

Proof:

$$\text{Suppose } H(x, y) = (x + 1, 2 - y).$$

We must show whether H is surjective or not.

Definition of surjective:

Outputs cannot go unused

Let $(m, n) \in \mathbb{R} \times \mathbb{R}$

Suppose $H(x, y) = (m, n)$, then $(x + 1, 2 - y) = (m, n)$.

Hence, $m = x + 1$, and $n = 2 - y$

$$x = m - 1$$

$$y = 2 - n$$

Since $H(x, y) = (x + 1, 2 - y)$, then:

$$H(m - 1, 2 - n) = ((m - 1) + 1, 2 - (2 - n)) = (m, n)$$

Therefore, $H(x, y) = (m, n)$, which is what was to be
shown.

□

Solution: H is surjective.